

Segway-like device gives mobility to paraplegics

For those with spinal cord injuries and illnesses, scientists in Istanbul, Turkey, have created a robotic mobilisation device that provides a level of independence far beyond that available to users of conventional wheelchairs.

The Tek device allows physically challenged people a range of locomotor movements such as moving around independently while being able to use their hands, stand up, squat and a host of other movements that are crucial to everyday life.

The device is entered from the back – unlike the front as with a wheelchair. By attaching a thick, padded strap around the hips, paraplegics can manoeuvre themselves into the device on their own. The device uses a suspension system that balances their weight so that they can stand up with just a gentle pull.

The platform, designed by Turkish R&D company AMS Mekatronik, comes in five sizes and is being marketed in Turkey at a price of \$15,000.

Robotic gloves for workers

A human grasp-assist device called Robo-Glove is being developed to help auto-workers and astronauts do their jobs better, while reducing the risk of injuries caused by constant physical stress.

Research shows that continuously gripping a tool can cause fatigue in hand muscles within a few minutes. Initial testing of the Robo-Glove indicates the wearer can hold a grip longer and more comfortably, reducing the amount of force factory workers need to hold a tool from 6.8 to 9.1 kg (15 to 20 pounds) to just 2.3 to 4.5 kg (5 to 10 pounds). It is expected to cut down the risk of repetitive motion injuries.

Actuators embedded in the upper portion of the glove provide grasping support to human fingers. Pressure



Robotic gloves for workers

sensors in the fingertips detect when the user is grasping a tool. When the user grasps the tool, the synthetic tendons automatically retract, pulling the fingers into a gripping position and holding them there until the sensor gives the signal to release.

Current prototypes of the Robo-Glove weigh about 1 kg, and include the control electronics, actuators and a small display for programming and diagnostics. An off-the-shelf lithium-ion power-tool battery with a belt-clip is used to power the system. A third-generation prototype that will use repackaged components to reduce the size and weight of the system is nearing completion.

Robotic jellyfish powered by hydrogen in water

Researchers at Virginia Tech have built Robojelly – a robotic jellyfish that runs by sucking up hydrogen while moving through water. The robot, still in the

early stages of development, can eventually be used in underwater search-and-rescue operations. Being fuelled by hydrogen underwater means the robot will not require an external power source and can also act as a vehicle.

Robojelly is built from shape-memory alloys. The robot is powered by heat produced from chemical reactions between oxygen and hydrogen in the water and the platinum powder. The heat from the reactions is transferred to the artificial muscles of the robot, reshaping them. Eight moving segments wrapped in carbon nanotubes and coated with a platinum powder replicate the natural opening-and-closing propulsion method of jellyfish.

The researchers' next step is to figure out a way to deliver hydrogen to each segment separately. This will allow them to be controlled individually so that the robot can move in different directions.